Discrete Time Markov Chains

Stochastic Models

May want to capture how a system evolves, using *dependent* random variables, described by some dynamics. A very simple model that sees wide application is a Discrete Time Markov Chain. They have been used for the following (this is far from exhaustive):

user web page navigation, cache contents and performance, speech recognition, machine learning (reinforcement learning), algorithmic music composition, sports

Introductory Example

Consider the following (very simplified) scenario:

- three web pages that link to each other *A, B, C*
- page *A* has links to *B* and *C*
- page *B* has links to *C*
- page *C* has links to *A* and *B*
- when a user leaves *A*, 1/3 of the time a link to *B* is selected, otherwise a link to *C* is selected
- *B* always links to *C*
- *C* links to *A* or *B* with equal probability
- Choice of next page is completely independent of the past (only depends on current page)

Q: What proportion of visits are made to each page?

Discrete-time systems

- this is a discrete-time system
- the *n*th “time point” corresponds to the *n*th page visited
- natural for some time settings: a continuous-time framework may be more suitable for others

Discrete-Time Markov Chains

Definition. A DTMC (discrete-time Markov chain) is a stochastic process $\{X_n, n = 0,1,2, \dots\}$ where X_n denotes the state at (discrete) time step n and such that, $\forall n \geq 0, \forall i, j$ and $\forall i_0, \dots, i_{n-1}$

$$P\{X_{n+1=j} | X_n = i, X_{n-1} = i_{n-1}, \dots X_0 = i_0\}$$
$$= P\{X_{n+1=j} | X_n = i\} = P_{ij}$$

- last equality is a property known as *stationarity:* statistics of process independent of time, P_{ij} is independent of the time step and of history
- "chain" refers to the state space being discrete

Markovian Property and P

- Markovian property – the conditional distribution of the next state X_{n+1}, given past states $X_0, X_1, \ldots, X_{n-1}$ and the present state X_n, is independent of past states and depends only on the present state X_n
- Transition (probability) matrix, P – associated with any DTMC is a matrix, P, whose i, j entry, P_{ij} represents the probability of moving to state j on the next transition, given that the current state is i
- P may have infinite dimension
- P is square

Markovian Property and P

- Rows of P sum to one – the system must be in some state at each step
- $P_{ii} > 0$ is possible

First Example Revisited

If the states are taken as *A, B, C* in that order, then

$$P = \begin{bmatrix} 0 & 1/3 & 2/3 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{bmatrix}$$

Umbrella Problem

- an absent-minded professor has two umbrellas that are used for making trips from home to work and back
- if it rains and an umbrella is at the current location, an umbrella is taken
- if it is not raining, no umbrella is taken
- it rains with probability p when each trip is made, independent of prior trips
- eventual goal: determine the fraction of trips during which the professor gets wet

Umbrella Problem (cont'd)

- state space: track the number of umbrellas at *current location*
- state transition diagram given on the board
- transition probability matrix is

$$P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1-p & p \\ 1-p & p & 0 \end{bmatrix}$$

Umbrella Problem (cont'd)

Could model this with six states, where the state is the ordered pair that consists of the number of umbrellas at home and the current location of the professor.

More examples

There are two more examples in the book, a repair model and a program analysis model – please try to read!

n-step Transition Probabilities

- Let $P^n = P \times P \times \cdots \times P$, i.e., n copies of P multiplied together
- P_{ij}^n: (i, j) entry of P^n
- What does P_{ij}^n represent?

2-step Transition Probabilities

- Let's look at a 3-state DTMC (states 0, 1 and 2)
- The probability of going from state 0 to state 1 in 2 steps is, by exhausting all possibilities ($0 \to 0 \to 1, 0 \to 1 \to 1, 0 \to 2 \to 1$):

$$P_{00}P_{01} + P_{01}P_{11} + P_{02}P_{21}$$

So, the required probability is P_{01}^2, the (0,1) entry of P^2. In general,

$$P_{ij}^2 = \sum_{k=0}^{M-1} P_{ik}P_{kj}$$

so the 2-step probabilities are given by the entries in P^2

n-step Transition Probabilities

In general, if we know that the $(n-1)$-step transition probabilities are given by P_{ik}^{n-1}, then by the exhaustive enumeration of all possibilities:

$$P_{ij}^{n} = \sum_{k=0}^{M-1} P_{ik}^{n-1} P_{kj}$$

This allows an inductive argument that the n-step transition probability of going from state i to state j is given by the (i, j) entry of P^n.

Limiting Probabilities

Let's return to the first example:

$$P = \begin{bmatrix} 0 & 1/3 & 2/3 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{bmatrix}$$

$$P^2 = \begin{bmatrix} 1/3 & 1/3 & 1/3 \\ 1/2 & 1/2 & 0 \\ 0 & 1/6 & 5/6 \end{bmatrix}$$

Limiting Probabilities

$$P^5 = \begin{bmatrix} .1944 & .2870 & .5185 \\ .0833 & .2222 & .6944 \\ .3472 & .3750 & .2778 \end{bmatrix}$$

$$P^{20} = \begin{bmatrix} .2309 & .3077 & .4614 \\ .2312 & .3079 & .4609 \\ .2305 & .3075 & .4620 \end{bmatrix}$$

It appears that all of the rows are becoming the same. What this says is that the probability of being in state j becomes independent of the starting state i.

Limiting Probabilities

Definition. Let

$$\boldsymbol{\pi_j} = \lim_{\boldsymbol{n}\to\infty} \boldsymbol{P_{ij}^n}$$

The values π_j represent the limiting probability that the DTMC is in state j. For an M-state DTMC, with states $0,1,\dots,M-1$

$$\pi = [\pi_0, \pi_1, \dots, \pi_{M-1}],$$

with $\sum_{i=0}^{M-1} \pi_i = 1$ is a vector of the limiting probabilities of being in each state.

Limiting Probabilities

- not obvious that limit always exists (in fact, sometimes it does not)
- not obvious that limiting probabilities form a distribution
- for our purposes, we will continue with these assumptions (they are not unreasonable)

Umbrella Problem Revisited

Here, with $p = 0.4$,

$$P^{30} = \begin{bmatrix} .230 & .385 & .385 \\ .230 & .385 & .385 \\ .230 & .385 & .385 \end{bmatrix}$$

So, one could conclude that the limiting probability that one gets wet is $p\pi_0 = (0.4)(0.23) = 0.092$.

Calculating Limiting Probabilities

Is there a better way to find π_i, as opposed to raising P to a large number?

Stationary Probabilities

Definition. A probability distribution π is said to be *stationary* for a DTMC if

$$\boldsymbol{\pi P} = \boldsymbol{\pi}$$

and

$$\sum_{i=0}^{M-1} \pi_i = 1$$

We refer to this set of equations as the *stationary equations*.

Stationary Probabilities

We can rewrite the equations as

$$\sum_{i=0}^{M-1} \pi_i P_{ij} = \pi_j, \qquad j = 0, \dots, M-1$$

and

$$\sum_{i=0}^{M-1} \pi_i = 1$$

In the first set of equations, the LHS determines the probability of being in state j if the initial distribution is π. Hence the term stationary, if we start with this distribution, it never changes.

Key Theorem

Theorem. Given a finite-state DTMC with M states, let

$$\boldsymbol{\pi_j} = \lim_{\boldsymbol{n}\to\infty} \boldsymbol{P_{ij}^n} > \mathbf{0}$$

be the limiting probability of being in state j and let $\pi = [\pi_{0,}\pi_1, \dots, \pi_{M-1}]$, where $\sum_{i=0}^{M-1} \pi_i = 1$ be the limiting distribution. Assuming that the limiting distribution exists, then π is also a stationary distribution and no other stationary distribution exists.

Proof

First we show that a limiting distribution satisfies the stationary equations.

$$\pi_j = \lim_{n\to\infty} P_{ij}^{n+1}$$

$$= \lim_{n\to\infty} \sum_{k=0}^{M-1} P_{ik}^{n} P_{kj}$$

$$= \sum_{k=0}^{M-1} \lim_{n\to\infty} P_{ik}^{n} P_{kj}$$

$$= \sum_{k=0}^{M-1} \pi_k P_{kj}$$

Proof

Next we show that *any* solution to the stationary equations must be equal to the limiting distribution. Suppose that π' is a stationary distribution and we start with initial state having distribution π'. So,

$$\pi'_j = P\{X_n = j\}, \forall n$$

$$= \sum_{i=0}^{M-1} P\{X_n = j | X_0 = i\} P\{X_0 = i\}, \forall n$$

$$= \sum_{i=0}^{M-1} P_{ij}^n \pi'_i, \forall n$$

Proof

$$\pi_j' = \lim_{n\to\infty} \pi_j'$$

$$= \sum_{i=0}^{M-1} \lim_{n\to\infty} P_{ij}^n \pi_i'$$

$$= \sum_{i=0}^{M-1} \pi_j \pi_i'$$

$$= \pi_j \sum_{i=0}^{M-1} \pi_i'$$

$$= \pi_j$$

Example

A computer program's execution is modelled as a DTMC. The state is the resource being used at time n. Suppose that there are three resources: CPU (state 0), Disk A (state 1), and Disk B (state 2). The probability transition matrix is

$$P = \begin{bmatrix} 0.9 & 0.05 & 0.05 \\ 0.2 & 0.8 & 0 \\ 0.2 & 0 & 0.8 \end{bmatrix}$$

1. What is the probability that, given the program is initially using the CPU, it is using Disk A two time units later?
2. What is the steady-state probability that Disk A is being used?

Example

Consider a multiprocessor system with two processors and two memory modules (MMs). At each time point, requests for an MM are made in the following manner.

1. If both processors are accessing different MMs, at the next time point each processor makes independent requests for the memory modules: with probability 0.4, a request is made for MM1, otherwise the request is made for MM2.
2. If both processors are accessing the same MM, only one processor makes a new request at the next time step (the other maintains its request for the same MM). The request probabilities are the same as in the first case.

Calculate the steady-state probability that the processors are accessing different memory modules.

Expected time to reach a state

- Consider an example with states $\{0,1,2,3\}$ and transition matrix

$$P = \begin{bmatrix} 0 & 0.4 & 0.6 & 0 \\ 0 & 0.4 & 0.6 & 0 \\ 0 & 0.9 & 0 & 0.1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Define $T_{0,3}$ to be the number of steps to enter state 3, given that the system starts in state 0.
- We can write

$$E\left[T_{0,3}\right] = 1 + 0.4E\left[T_{1,3}\right] + 0.6E\left[T_{2,3}\right]$$

Expected time to reach a state

- On the RHS, the value comes from the fact that we always take one step from 0, no matter what. We then need to add the number of steps starting from the next state. With probability 0.4, we count the number of steps from state 1, and with probability 0.6, the number of steps from state 2.
- In a similar manner, we can generate the additional equations:

$$E\left[T_{1,3}\right] = 1 + 0.4E\left[T_{1,3}\right] + 0.6E\left[T_{2,3}\right]$$
$$E\left[T_{2,3}\right] = 1 + 0.9E\left[T_{1,3}\right] + 0.1E\left[T_{3,3}\right]$$

Expected time to reach a state

- Using $E[T_{3,3}] = 0$, we now have three equations in three unknowns.
- Solve these to yield $E\left[T_{0,3}\right] = 26.67$

General Result

- Assume states are $\{0,1,\dots,N\}$
- Assume that we are interested in computing $E\left[T_{0,N}\right]$.
- For $j = 0,\dots,N-1$, generate

$$E\left[T_{j,N}\right] = 1 + \sum_{l=0}^{N} P_{jl} E\left[T_{l,N}\right]$$

- Set $E\left[T_{N,N}\right] = 0$, wherever it appears.
- Solve the resulting set of linear equations.

Caching Example

Introduction

- wish to compare strategies for virtual memory paging in OS
- will model the system as a *ranked* set of pages
- suppose that we have eight pages, currently ranked $P_3, P_7, P_2, P_6, P_5, P_1, P_8, P_4$
- next request for page P_5
- new ranking under LRU?
- new ranking under Move-Ahead (move up one in the ranking)?

Introduction

- updated rank for LRU: $P_5, P_3, P_7, P_2, P_6, P_1, P_8, P_4$
- updated rank for Move-Ahead: $P_3, P_7, P_2, P_5, P_6, P_1, P_8, P_4$
- suppose that the first three pages in memory, cache miss if not in memory

System State

- if N is the number of pages, state is permutation of $\{1, \ldots, N\}$
- size of the state space is $N!$
- Linux system: page size 4 kB, memory size 1 GB, swap disk size 1 GB, so state space has size $500000!$

Request Model

- requests random
- requests follow an i.i.d. (independent and identically distributed) sequence with $P\{R_n = P_i\} = p_i$
- suppose that we want to consider performance when there is a "more frequent page" P_A
- assume that

$$a = P\{R_n = P_A\}$$
$$b = P\{R_n = P_i\},\ P_i \neq P_A$$

- a and b satisfy $a > b$ and $a + (N-1)b = 1$.

Request Model

- note that this allows that a is (much) smaller than $(N-1)b$
- can we reduce the state space?

State Space Reduction

- $\{X_n\}$ is the position of page P_A after the nth request
- state space is size N
- assume $N = 8$

LRU: Transition Matrix

$$P = \begin{bmatrix} a & 7b & 0 & 0 & 0 & 0 & 0 & 0 \\ a & b & 6b & 0 & 0 & 0 & 0 & 0 \\ a & 0 & 2b & 5b & 0 & 0 & 0 & 0 \\ a & 0 & 0 & 3b & 4b & 0 & 0 & 0 \\ a & 0 & 0 & 0 & 4b & 3b & 0 & 0 \\ a & 0 & 0 & 0 & 0 & 5b & 2b & 0 \\ a & 0 & 0 & 0 & 0 & 0 & 6b & b \\ a & 0 & 0 & 0 & 0 & 0 & 0 & 7b \end{bmatrix}$$

LRU: Stationary Distribution

With $a = 0.3, b = 0.1,$

$$\pi^T = \begin{bmatrix} 0.30 \\ 0.23 \\ 0.18 \\ 0.12 \\ 0.08 \\ 0.05 \\ 0.03 \\ 0.01 \end{bmatrix}$$

Move-Ahead: Transition Matrix

$$P = \begin{bmatrix} a+6b & b & 0 & 0 & 0 & 0 & 0 & 0 \\ a & 6b & b & 0 & 0 & 0 & 0 & 0 \\ 0 & a & 6b & b & 0 & 0 & 0 & 0 \\ 0 & 0 & a & 6b & b & 0 & 0 & 0 \\ 0 & 0 & 0 & a & 6b & b & 0 & 0 \\ 0 & 0 & 0 & 0 & a & 6b & b & 0 \\ 0 & 0 & 0 & 0 & 0 & a & 6b & b \\ 0 & 0 & 0 & 0 & 0 & 0 & a & 7b \end{bmatrix}$$

Move-Ahead: Stationary Distribution

With $a = 0.3, b = 0.1,$

$$\pi^T = \begin{bmatrix} 0.67 \\ 0.22 \\ 0.07 \\ 0.02 \\ 0.01 \\ 0.01 \\ 0.00 \\ 0.00 \end{bmatrix}$$

Conditional Cache Miss Probabilities (for Page A)

Cache Size	LRU	Move-Ahead
0	1.00	1.00
1	0.70	0.33
2	0.47	0.11
3	0.28	0.04
4	0.17	0.02
5	0.09	0.01
6	0.04	0.00
7	0.01	0.00
8	0.00	0.00